

Session 5: Subagents, Hooks, and Final Project

Spawning focused agents, event-driven automation, and an end-to-end research artifact

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Start-of-Session Ritual

Same three steps as last session, with today's folder.

```
# 1. Pull the latest course repo
cd ~/claude-code-workshop && git pull

# 2. Copy today's session folder into your workspace
cp -r session_5/ ~/my_workspace/session_5/

# 3. Open your copy in VS Code and work there
cd ~/my_workspace/session_5 && code .
```

Don't have the course repo yet?

```
git clone https://github.com/bradyallardice/claude-code-workshop.git
```

Five minutes, together, live. Everyone in a clean state before we move on.

If `git pull` fights back, use the recovery from Session 4: `git stash -u`, `git branch backup-s5`, `git fetch origin`, `git reset --hard origin/student`, `git stash pop`.

Today: Agents, Hooks, and a Paper

Three parts.

- 1 **Agents.** Subagents, named specialists, and multi-agent orchestration — when to spawn one, how to define your own, how they compose.
- 2 **Hooks.** Shell commands that run automatically on events (file writes, tool use, session start) — enforcement that doesn't depend on Claude remembering.
- 3 **Putting it together.** Start from an unfamiliar paper, use a Zotero library to write the literature review, then build a short replication + extension note.

The capstone is not a test of whether a prepared AI can finish a template. It is a test of whether you can learn the paper, direct the workflow, verify the output, and route the work.

A focused agent spawned for a bounded job —
the right tool for the parallelizable, scoped work
that already shows up in your pipeline.

Concrete patterns, then named teams.

What a Subagent Is

Claude Code can spawn another Claude Code instance inside your current session — a subagent.

- It has its own context window
- It gets a scoped task and a scoped tool set
- It returns a summary to your main conversation — not its raw trace

What that buys you:

- ➊ **Parallelism** — run three searches at once instead of one after the other
- ➋ **Context hygiene** — a big messy exploration stays out of your main context budget

Think of it as

Delegating a bounded sub-task to a focused colleague who returns a clean, structured answer — so your main conversation stays on the high-level work.

Where Subagents Shine

Reach for a subagent when the subtask is:

- **Parallelizable** — independent work that doesn't need to be sequenced
- **Bounded** — clear inputs, clear outputs, no back-and-forth
- **Independently verifiable** — you can check the result without reading the full trace

Concrete research use cases:

- **Multi-database literature searches** — one agent per database, all running in parallel
- **Coding many documents** against a fixed rubric
- **Parallel robustness checks** across independent specifications
- **Running the same sanity-check** script over many datasets
- **Exploring a codebase** — send a subagent to map out the files, keep the summary in your main context

How to Trigger a Subagent

You don't have to set anything up — just ask.

Claude Code ships with a general-purpose subagent built in. To spawn one, describe the work in plain English:

- *“Send a subagent to map the directory structure”*
- *“Use a subagent to find every place we set the random seed”*

For parallelism, say so explicitly.

- *“Spawn three agents **in parallel** to search Zotero, Google Scholar, and Connected Papers”*
- “In parallel” / “at the same time” / “concurrently” all signal that the agents should run in one batch rather than sequentially

No setup required

The general-purpose agent ships with Claude Code. Defining your own named agent (later this session) is for when you want to reuse a scoped role across many sessions.

Using a Generic Subagent in Practice

Worked example: one subagent, one bounded local task.

Your prompt to Claude:

```
Spawn a subagent to read the slides for sessions 1-4
and return a deduplicated list of every unique terminal
command I'll need across the course.
```

```
Output as: command | what it does | first session
it appears in.
```

What Claude does:

- Spawns one general-purpose subagent and hands it the four .tex files
- The subagent does the reading, extraction, and dedup in its own context
- You get back the table; the slide text never lands in your main conversation

Three habits that make subagents pay off:

- ❶ **Brief them like a contractor.** Concrete deliverable, all the context they need up front — they don't see your conversation history.
- ❷ **Ask for structured output** (bullet list, table, JSON). You'll be consuming the result; make it easy to read and verify.
- ❸ **Verify independently.** You get a summary, not the full trace. Spot-check the way you would a student's work.

The payoff

A literature sweep, robustness battery, or repo survey that would have taken an hour finishes in minutes — and doesn't eat into your main conversation's context. That's what makes them worth reaching for.

What a Predefined Subagent Looks Like

A small markdown file — author it once, Claude spawns it many times.

Where it lives: `~/.claude/agents/<name>.md` (user-level) or
`.claude/agents/<name>.md` (project-level).

What you specify (metadata + body):

- **Identity** — name and a one-line description. The description is what Claude reads to decide when to spawn this agent.
- **Tool allowlist** — exactly which tools it can use (e.g. Read only, no Bash).
- **Model** — Opus, Sonnet, or Haiku.
- **System prompt** — the body of the file: role, disposition, expected output format.

Plain text — metadata sits between two `---` lines at the top of the file; the system prompt is everything below.

Building a Specialized Agent

Worked example: turn “audit my regression script” into a reusable role.

Walking through the four fields:

- 1 **Name + description.** What it does, plus the one-line trigger Claude reads to decide when to spawn it.
“Audit a regression script for silent issues.”
- 2 **Tool allowlist.** Read, Grep, Glob — read-only. A critic that can fix code is no longer a critic.
- 3 **Model.** haiku — fast and cheap, fine for systematic pattern-matching.
- 4 **System prompt.** Disposition + output contract: *“You’re a skeptical reviewer. Flag invalidating issues. Return: line, issue, why. Don’t propose fixes.”*

Save as `~/.claude/agents/spec-critic.md`.

Now in any session:

- Claude auto-spawns it when a task matches the description
- Or invoke explicitly: *“use spec-critic on solution_estimate.py”*
- Role is enforced — if it suggests a fix, it broke its own contract

Permissions still apply: tools the agent uses must be allowed in `.claude/settings.json`.

Why Define a Named Subagent at All?

If skills are reusable, why not just spawn ad-hoc subagents that pick up the right skill?

The framing

Skills are knowledge. Subagents are roles.

A role bundles a disposition, a tool scope, and an invocation contract — and then draws on skills to execute.

Three things a role gives you that a skill can't:

- ❶ **Tool scope — enforced.** A code-review subagent has read-only access, no bash. Skills are advisory; tool scopes aren't.
- ❷ **Identity, not advice.** The system prompt frames the whole disposition — “find problems, don't fix them.” Subagents *are* the disposition.
- ❸ **A stable invocation contract.** Delegate code review → get a structured critique. Ad-hoc means re-specifying the contract each time.

Pre-define a subagent when the **role** is the reusable unit, not just the know-how.

When one specialist isn't enough.
Compose subagents into pipelines, fan-outs,
and critic-builder loops.

Three patterns, one worked example.

Three Composition Patterns

Most multi-agent workflows are one of three shapes.

- ① **Fan-out / map-reduce.** N inputs, one specialist per input, merge at the end.
Example: code 50 interview transcripts against the same rubric — one agent per transcript, then synthesize.
- ② **Pipeline.** Each stage's output is the next stage's input. Named and generic agents can mix.
Example: spec-critic flags issues → generic agent drafts fixes → spec-critic re-audits.
- ③ **Critic-builder loop.** One agent produces, another reviews, the producer revises.
Example: a paragraph drafter ↔ a paragraph-reviewer, two or three rounds.

Worked example: hardening a regression script in three stages.

Your prompt to Claude:

Run this pipeline on `solution_estimate.py`:

1. Use `spec-critic` to list every silent issue.
2. Spawn a subagent to draft minimal-diff fixes for each issue.
3. Re-run `spec-critic` on the patched script.

Return: original issues | proposed fixes | residual issues.

What makes this work:

- Each stage has a stable input/output contract — so they compose
- Two `spec-critic` calls share a role but run in fresh contexts — consistent without state
- Your main conversation only sees the final summary; intermediate traces stay isolated

One Agent or Many?

Default to one agent. A team has to earn its keep.

Stick with one agent when:

- The task is bounded enough to fit one context window
- One disposition is enough — no producer/critic split
- You're still exploring — you don't yet know the right stages
- Orchestration overhead would dominate the actual work

Reach for a team when:

- The work has naturally distinct roles (writer + reviewer; auditor + fixer)
- Independent verification is the point — a separate set of eyes
- You'll run the workflow many times — the setup cost amortizes
- N independent inputs need the same treatment in parallel

The test

Each extra agent is a stage that can drift. Add one only if it does something the previous one demonstrably can't.

Shell commands the harness runs automatically
on events — file writes, tool calls, session start.
Enforcement that doesn't depend on Claude remembering.

Concept, one example, when to reach for one.

What a Hook Is

A hook is a shell command wired to a Claude Code event. Configured in `.claude/settings.json`; run by the harness, not by Claude.

Common events:

- **PreToolUse** — before a tool runs (can block it)
- **PostToolUse** — after a tool runs
- **SessionStart** — at the start of a session
- **UserPromptSubmit** — when you submit a prompt
- **Stop** — when Claude finishes responding

What this buys you:

- **Enforcement, not advice.** The hook runs whether Claude remembers it or not.
- **Auditability.** Log every tool use, every edit, every Bash call.
- **Automation.** Format on save, refresh derived files, notify on completion.

Worked Example: A Research Audit Log

Goal: every Bash command Claude runs gets logged with a timestamp — automatically, every session.

Add to `.claude/settings.json`:

```
{
  "hooks": {
    "PreToolUse": [{
      "matcher": "Bash",
      "hooks": [{
        "type": "command",
        "command": "echo \"$(date '+%F %T') $CLAUDE_TOOL_INPUT\" >> ~/research_log.txt"
      }]
    }]
  }
}
```

What happens: Before every Bash call, the harness runs the command above and appends a timestamped line to your research log. Claude can't bypass it — the harness, not the model, runs hooks.

Installing a Pre-Built Hook: `claude-ignore`

Many useful hooks already exist — you don't have to write your own.

What `claude-ignore` does:

- A `PreToolUse` hook on `Read` that blocks files matched by a `.claudeignore` (gitignore-syntax patterns; comments and blank lines are ignored)
- Walks up from the current directory to root, merging every `.claudeignore` it finds — different rules at different folder levels
- On a match, exits with code 2: hard block. Claude sees the failure and can't bypass it
- Use for `.env`, `*.secret`, raw data — anything you don't want in the model's context

Install + initialize:

```
npm install -g claudeignore  
cd ~/my_workspace/my_project && claude-ignore init
```

`init` creates a `.claudeignore` and wires the hook into `.claude/settings.json`.

Source: github.com/li-zhixin/claude-ignore. Requires Node.js (npm).

Hook, Skill, or Agent?

Three layers of automation, three different things.

The framing

Skills are knowledge. Subagents are roles. Hooks are guarantees.

Reach for a:

- **Skill** when Claude needs to know *how* to do X.
- **Subagent** when you want a delegated worker with a scoped role.
- **Hook** when something must happen regardless of what Claude remembers.

Hooks are the only one of the three the harness enforces. Skills are advice; subagents have agency; hooks are mechanical.

Read an unfamiliar paper,
write a short literature review,
replicate one published figure,
run one small extension.

The point

Use Claude Code as a research assistant, not as a vending machine. You learn the paper, direct the task, inspect the code, verify the output, and decide what the result means.

The first output is not code. It is orientation.

Before asking Claude to replicate anything, you should be able to answer:

- What is the paper's research question?
- What is the paper's core contribution?
- What treatment, attributes, and outcome are being analyzed?
- What does Figure 1 estimate, and why is it central?
- What does the assigned extension change or add?
- Which literatures does the paper claim to speak to?

Read the paper and my notes first. If Claude starts coding before you understand the target, stop it.

Step 1: Clone, Teams, Branches

Two-person teams, all working in the GitHub collaboration repo from Session 4.

```
git clone https://github.com/bradyallardice/github_collaboration_practice.git
cd github_collaboration_practice
git switch -c team-<n>/<your-name>-<strand>
```

Assign two ownership lanes before touching files:

- **Partner A: paper + literature lane** — paper notes, Zotero search, literature review draft, claim checking
- **Partner B: data + analysis lane** — data validation, Figure 1 script, extension script, output checks
- **Both: reciprocal review** — each person reviews the other's PR before merging into the team branch

Everyone is doing the same capstone task. Branches keep identical work from colliding. Reviews keep partner specialization from becoming blind trust.

Step 2: Zotero, Not Local BibTeX

The literature review uses a prebuilt Zotero library.

The workflow:

- ❶ Import the provided BibTeX seed into Zotero.
- ❷ Confirm the Zotero MCP can find the papers.
- ❸ Remove the local import file from the project workspace.
- ❹ Ask Claude to search Zotero, not to mine a local `.bib` file.

Why

A local `.bib` file lets Claude pattern-match citations without understanding the literature. Zotero forces the workflow closer to the thing you would do in a real project: search, inspect, cite, and verify.

Instructor setup: the seed file should be a one-time import file, not a source Claude reads during drafting. If it must sit in the repo, block it with `.claudeignore` and the read guard.

Step 3: Write the Literature Review

Use Claude for search and synthesis, but keep the claims accountable.

A useful first prompt:

Search the Zotero library for papers most relevant to this article's research question. Return a table with citation key, mechanism, empirical setting, and how it connects to the capstone paper. Do not write prose yet.

Then:

- Open the most important papers or notes yourself.
- Decide which claims belong in a short review.
- Draft 2–3 paragraphs that position the capstone paper.
- Check that every citation supports the sentence attached to it.

What Is Provided

Course infrastructure

- A template repo with data, starter scripts, paper skeleton
- The paper, instructor notes, and a Zotero library seed
- Hooks that protect `reference_code/`, `main`, and ignored files
- General skills: data validation, robustness, tables, references, prose review
- Reviewer subagents: code critic, results trace, PR referee, note reviewer

Your work

- Read the paper and instructor notes
- Write the literature review from the Zotero library
- Write the replication instruction yourself
- Decide what Claude may infer and what you must specify
- Read the code and check the output
- Interpret the extension cautiously
- Explain which AI workflow pieces would transfer to your own work

The safety rails are prebuilt. The analytical direction is not.

Why Figure 1 Is Not a Skill

Inside this capstone, reproducing Figure 1 happens once.

It is project-specific: one paper, one dataset, one reference figure, one visual target.

So it should not be an installed skill. The transferable move is not “run my Figure 1 skill.”
The transferable move is:

- ① Read the reference code
- ② State the reshape target and estimand
- ③ Specify what the output should look like
- ④ Tell Claude what checks to print before saving
- ⑤ Review the plan, the code, and the final figure

The routing lesson

Some work is one-off for this project but still teaches a reusable pattern for your own research.

Your Replication Prompt

Before Claude writes code, write the instruction yourself.

It should specify:

- **Input:** `data/clean_AJPS.csv`
- **Reshape:** one respondent-task per row
- **Estimand:** marginal means by AI vs. Offshoring treatment
- **Visual target:** four stacked panels matching Figure 1
- **Output:** `paper/figures/figure1_replication.png`
- **Checks:** row counts, task counts, treatment counts before save

Ask Claude for a plan first. If the plan chooses a join, changes the sample, or skips verification, reject it.

- ➊ **Clone and organize.** Two-person teams in `github_collaboration_practice`; one branch per person.
- ➋ **Set the citation boundary.** Import the seed into Zotero; remove or block the local `.bib`.
- ➌ **Understand the paper.** Read the article and instructor notes.
- ➍ **Write the literature review.** Use Zotero search, then verify the claims.
- ➎ **Validate the data.** Use `spec-validator`; decide what any warning means.
- ➏ **Replicate Figure 1.** Prompt, plan, code, run, inspect, commit.
- ➐ **Run the extension and finish the note.** Verify sample, estimand, prose, and tables.
- ➑ **Reflect.** Fill in `AI_WORKFLOW_REFLECTION.md`.

The deliverable is not just the PDF. It is the PDF plus evidence that you stayed in control of the workflow.

The Reflection Is the Transfer

The capstone uses course-provided infrastructure. Your own research will not.

In `AI_WORKFLOW_REFLECTION.md`, answer:

- What did you delegate to Claude?
- What did you personally verify?
- What did you refuse to let Claude decide?
- What would belong in `CLAUDE.md` for your own paper?
- What repeated convention would become a skill?
- What reviewer role would become a subagent?
- What safety rule would become a hook?

What I am looking for

Not “I used all the tools.” I want to see that you can choose the right layer for the next project.